

Amarnath D.J.

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SUMMARY

B.Tech Computer Science (AI & ML) student at SCTCE with hands-on experience building end-to-end machine learning pipelines, deploying production-grade web applications, and applying deep learning and computer vision techniques to real-world problems. Proficient in Python, Scikit-learn, TensorFlow, XGBoost, and Flask. Actively building expertise in data analysis, predictive modeling, and MLOps. Seeking roles in AI/ML Engineering, Data Science, Data Analytics, or Full-Stack Development.

SKILLS

AI & ML: Scikit-learn, TensorFlow, XGBoost, SARIMA, Regression, Classification, Clustering, Model Evaluation

Deep Learning & Computer Vision: YOLOv8, ByteTrack, LSTM, CNN, OpenCV, Object Detection, Object Tracking

Data Science & Analysis: Pandas, NumPy, EDA, Data Wrangling, Statistical Analysis, Data Cleaning

Data Visualization: Matplotlib, Seaborn, Chart.js, Plotly

Languages: Python, Java, C, SQL, HTML

MLOps & Deployment: Flask, FastAPI, Streamlit, Docker, Render, AWS (basic), Azure (basic)

Tools: GitHub, Jupyter, VSCode, Notion

Data Structures & Algorithms: Arrays, Linked Lists, Stacks, Queues, Search and Sort Algorithms

Other: Object Oriented Programming, Operating Systems, MS Word, MS Excel

PROJECTS AND RESEARCH

ML-Based Acoustic Rain Gauge

In Progress

Tech: Audio Processing, Librosa, Machine Learning, Time Series

- Developing an acoustic rainfall estimation system using audio signal processing and machine learning.
- Predicting rainfall intensity directly from environmental audio recordings collected in real-world conditions.
- Synchronising model output with mechanical rain gauge measurements for ground-truth validation.
- github.com/Amarnathdj10/ML-based-Acoustic-Rain-Gauge

Vision-Based Crowd Footfall Prediction System

2025-26

Tech: YOLOv8, ByteTrack, SARIMA, OpenCV, Python

- Built a real-time crowd monitoring system using YOLOv8 and ByteTrack for people detection and tracking.
- Applied SARIMA time-series forecasting to predict crowd flow from CCTV-derived data.
- <https://github.com/Amarnathdj10/Vision-Based-Crowd-Footfall-Prediction-at-Choke-Point-Entries>

Pre-Match Player Rating Prediction using Machine Learning

2025-26

Tech: Python, XGBoost, Flask, Feature Engineering

- Built a full-stack ML application to predict football player match ratings using XGBoost and Flask.
- Engineered performance metrics including rolling form, opponent strength, and rest days.
- Deployed the model as a production-ready web application on cloud infrastructure.
- <https://github.com/Amarnathdj10/Pre-Match-Player-Rating-Prediction-using-Machine-Learning>

FIFA World Cup 2026 Predictor

In Progress

Tech: Python, Monte Carlo Simulation, Elo Ratings, Football Analytics, ML

- Building a tournament simulation and prediction engine for the 2026 FIFA World Cup.

- Incorporating team strength metrics, Elo ratings, squad valuation data, and Monte Carlo simulations.
- Estimating qualification probabilities and tournament winners through statistical modelling.
- github.com/Amarnathdj10/Fifa-World-Cup-2026-Predictor

EDUCATION

B.Tech in Computer Science – Artificial Intelligence & Machine Learning *Kerala*
Sree Chitra Thirunal College of Engineering (SCTCE), Thiruvananthapuram 2023 – Present

ICSE / ISC Board (Class X & XII) *Kerala*
Loyola School, Thiruvananthapuram 2008 – 2022

CERTIFICATIONS

- **Python for Data Science** (NPTEL) – IIT Madras
- **Reverse Engineering Bootcamp** (LEAP) – IIT Madras
- **Automation with Arduino Bootcamp** (LEAP) – IIT Madras

ACTIVITIES AND ACHIEVEMENTS

- KTU A-Zone Chess Championship 2025 – Runner-Up
- SCTCE Inter-Department Chess Championship – Champion, 2025
- SCTCE Chess Club Male Lead, 2025/26